`caffe2-module`

Crate in the process of being translated from C++ to Rust. Some function bodies may still be undergoing translation.

`ModuleSchema`

This type represents the schema of a Caffe2 module. A module is a collection of operators and associated state that can be run as a single unit. The schema of a module defines the inputs and outputs of the module, as well as any associated parameters or attributes.

`current_module_handles`

`current_modules`

`g_module_change_mutex`

`has_module`

`load_module`

`mutable_current_modules`

These functions and variables are used to manage the state of Caffe2 modules. `current_module_handles` and `current_modules` provide access to the current modules and their handles, respectively. `g_module_change_mutex` is a mutex used to synchronize access to the current module state. `has_module` checks if a module is currently loaded, while `load_module` loads a module into the current module state. `mutable_current_modules` provides a mutable reference to the current module state, which can be used to modify the state of the loaded modules.

`Caffe2ModuleTestStaticDummyOp`

This type represents a dummy operator used for testing Caffe2 modules. The operator does not perform any computation and is used only to test the loading and running of Caffe2 modules.

`caffe2_module`

`module_test_dynamic_module`

`module_test_static_module`

These types represent Caffe2 modules used for testing and validation. `caffe2_module` is a macro used to define a new Caffe2 module, while `module_test_dynamic_module` and `module_test_static_module` are specific modules used for testing and validation.

`register_cpu_operator`

This function registers a new CPU operator with Caffe2. An operator is a function that performs a specific computation, such as a convolution or a matrix multiplication. Registering a new operator with Caffe2 allows it to be used in Caffe2 modules and networks.

`run`

This function runs a Caffe2 module. Running a module executes all the operators in the module in the correct order, with the appropriate inputs and parameters.

`typename`

This function returns the name of the type of a given value. It is used to provide type information for Caffe2 modules and operators.

In summary, `caffe2-module` is a Rust crate that provides functions and types for managing and running Caffe2 modules. Caffe2 modules are collections of operators and associated state that can be run as a single unit, and are used in deep learning for tasks such as training and inference. The crate provides functions for loading and managing modules, as well as for registering new operators with Caffe2. The crate is still in the process of being translated from C++ to Rust, but many of the core functions are already available.